Chapter 55 — Widgets & App Shortcuts

> **Part VI — Lifecycle, Background & Platform** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 30 (WorkManager), Ch 3 (SSOT/Room), Ch 32 (sync), Ch 52 (deep links), Ch 47 (background limits), Ch 44 (lifecycle). Pairs with Ch 50 (notifications), Ch 42 (multi-account).
>
> **Reference surface in StreamCart:** surfaces *outside* the app — home-screen **widgets** (order status, cart total, continue-watching) built with Glance, and **app shortcuts** (static/dynamic/pinned long-press actions like "Search", "My Orders", "Resume last show") that deep-link into the app.
>
> **Note:** this chapter completes **Part VI — Lifecycle, Background & Platform (Ch 44–55)**.

55.1 Problem Definition and Real-World Context

Not everything happens inside your Activity. The **home screen** and **launcher** are app surfaces too: **widgets** show glanceable, periodically-updated content on the home screen, and **app shortcuts** give long-press quick actions on the launcher icon. Done well, they drive re-engagement and fast task access; done poorly, they show stale/wrong data, drain battery with over-frequent updates, leak content, or crash on the restricted execution model these surfaces run under.

Concretely, teams hit:

- **The widget execution model is restrictive.** A widget runs in the **launcher's process** via `RemoteViews` — you **cannot** run arbitrary UI code or hold a long-lived process; updates are **pushed** to it, subject to background limits (Ch 47). Treating a widget like an in-app screen fails. **Glance** (Compose-style widget API) still compiles down to `RemoteViews` with these constraints.
- **Stale or wrong data.** Widgets must reflect the **SSOT** (Ch 3) and update on data change / periodically (WorkManager, Ch 30) — not hold their own copy. Over-frequent updates waste battery and hit update-rate limits; under-updating shows stale info.
- **App shortcut types confused. Static** (manifest, fixed), **dynamic** (runtime-pushed, e.g., recent items), and **pinned** (user places on home screen) have different lifecycles and limits (typically ~4–5 visible). Using the wrong type, or not updating dynamic shortcuts, gives stale/empty menus.
- **Deep-link integration.** Both widgets and shortcuts must **deep-link** (Ch 52) into the right screen with the right back stack, auth gating, and account (Ch 42) — they're entry points like notifications (Ch 50).
- **Multi-account & privacy.** A widget on a shared/home screen can **leak** another account's or sensitive data; account switching (Ch 42) must update widgets; sensitive content needs care on an always-visible surface.
- **Lifecycle & updates under limits.** Scheduling widget refresh respecting Doze/App Standby (Ch 47), handling the widget being added/removed, and surviving process death.

The architecture: build widgets with **Glance** backed by the **SSOT** (Ch 3), updated by **WorkManager** (periodic + on-data-change, respecting Ch 47) and rendered as **`RemoteViews`** under the restricted model; define **app shortcuts** (static for fixed actions, **dynamic** for contextual/recent, pinned on user request) that **deep-link** (Ch 52) into the app;

integrate **deep links** for both surfaces (right destination, back stack, auth, account); and handle **multi-account/privacy** (update on switch, don't leak) and **lifecycle** (add/remove, update under background limits). This chapter is about **first-class app surfaces beyond the Activity, built within their constraints**.

Three practical use cases

1. **Order-status widget (StreamCart).** A home-screen widget shows the latest order's status from the SSOT (Ch 3), refreshed by WorkManager on sync (Ch 32) and periodically; tapping deep-links to the order (Ch 52).
2. **Continue-watching widget / shortcut.** A widget tile + a dynamic shortcut "Resume last show" that deep-links into the player at the saved position (Ch 58/59).
3. **Dynamic shortcuts for recent searches.** Long-pressing the icon shows recent searches/categories (dynamic shortcuts), updated as the user browses, each deep-linking to results.

Why this is architectural

Widgets/shortcuts are **out-of-process entry points** bound to the SSOT (Ch 3), background scheduling (Ch 30/47), deep linking (Ch 52), and multi-account/privacy (Ch 42/28). The decisions — SSOT-backed updates, update cadence under limits, shortcut types, deep-link integration, leak prevention — determine whether these surfaces are fresh, efficient, safe, and correct.

55.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Render glanceable widget content from the SSOT (Ch 3).
F2	Update widgets on data change + periodically, respecting background limits (Ch 47).
F3	Provide app shortcuts (static/dynamic/pinned) that deep-link into the app.
F4	Deep-link widget/shortcut taps to the right destination + back stack + auth (Ch 52).
F5	Handle multi-account (update on switch) and privacy (no leak) (Ch 42/28).
F6	Handle widget/shortcut lifecycle (add/remove/limits) and process death.

Non-functional requirements

Attribute	Policy
Freshness	Reflects SSOT; timely updates without staleness.
Battery	Update cadence respects Doze/App Standby (Ch 47); no over-refresh.
Correctness	Right data, right deep-link; works under the RemoteViews model.
Privacy	No sensitive/cross-account leak on an always-visible surface.
Resilience	Survives process death; handles add/remove gracefully.

Scope

- **In scope:** widgets (Glance + `RemoteViews` model, `AppWidgetProvider`), SSOT-backed content, WorkManager-driven updates (periodic + on-change) under Ch 47, app shortcuts (static/dynamic/pinned, limits), deep-link integration (Ch 52), multi-account/privacy (Ch 42/28), lifecycle (add/remove/update/process death).
- **Out of scope:** deep-link routing/security detail (Ch 52), WorkManager internals (Ch 30), Doze detail (Ch 47), SSOT internals (Ch 3), notifications (Ch 50), Compose UI (Ch 6 — Glance is similar but separate).

Assumptions & constraints

- Jetpack **Glance** for widgets (compiles to `RemoteViews`), `ShortcutManager` / `ShortcutManagerCompat` for shortcuts, WorkManager, Room, Hilt. `minSdk 24` (dynamic/ pinned shortcuts 25+; Glance has its own min). Widgets run out-of-process under background limits; shortcut counts are capped.

55.3 System Design Fundamentals

- **Widgets run out-of-process via `RemoteViews` (the key constraint).** A home-screen widget is hosted in the **launcher's** process; you don't get a live Activity. You build a limited view tree (`RemoteViews`) and **push** updates to it via `AppWidgetManager`. You **cannot** run arbitrary code continuously, hold state, or use normal Compose. **Glance** gives a Compose-like authoring API but **compiles to `RemoteViews`**, so the same limits apply (limited components, no arbitrary logic in the view, updates pushed from your process).
- **Widget data must come from the SSOT (Ch 3).** The widget renders a **snapshot** of app state. Keep the source of truth in Room/DataStore (Ch 3/4); when it changes (sync, Ch 32, or user action), **trigger a widget update** that reads the SSOT and pushes new `RemoteViews`. The widget never holds its own authoritative copy.
- **Updates are scheduled, under background limits (Ch 47):**
  - **Periodic:** `android:updatePeriodMillis` has a **30-minute minimum** and is unreliable under Doze — for anything finer/more reliable use **WorkManager** (Ch 30) with constraints (Ch 47) to refresh.
  - **On-change:** when the SSOT changes, enqueue a (possibly expedited) WorkManager update so the widget reflects it promptly.
  - **Respect limits:** don't over-update (battery + rate limits); batch; update only when data actually changed.
  - Glance offers `GlanceAppWidget.update(context, id)` and state via `updateAppWidgetState` to push new content.
- **App shortcuts (three types):**
  - **Static:** declared in XML (manifest `<meta-data>`), **fixed** at build time (e.g., "Search", "My Orders"). Always present.
  - **Dynamic:** pushed at **runtime** via `ShortcutManagerCompat.pushDynamicShortcut` (e.g., recent items, continue-watching). Update as context changes; **capped** (typically ~4–5 visible including static).
  - **Pinned:** the **user** places a shortcut on the home screen (`requestPinShortcut`); it persists until removed. Request only on user action.
  - All are long-press launcher actions that fire an **Intent** (a deep link, Ch 52).

- **Deep-link integration (Ch 52).** Widget taps and shortcut launches carry an **Intent/deep link** into the app → route through the **central router** (Ch 52) to the right destination with a synthesized back stack, auth gating (Ch 21), and account check (Ch 42). Treat the carried data as untrusted (Ch 52) and build `PendingIntent`s with `FLAG_IMMUTABLE`.
- **Multi-account & privacy (Ch 42/28).** A widget sits on a potentially shared surface:
  - **No sensitive leak:** avoid showing sensitive content (balances, private messages) on an always-visible widget unless the user opts in; consider a tap-to-reveal.
  - **Account correctness:** on account switch (Ch 42), **update/clear** widgets so they reflect the active account; don't show account A's order on account B.
- **Lifecycle (Ch 44/8):** handle widget **added** (`onEnabled`/`onUpdate`), **removed** (`onDisabled`/`onDeleted` — cancel its update work), **resize/reconfigure**, and **process death** (widget state is re-derived from the SSOT on next update, so nothing to restore — this is a benefit of SSOT-backing). Shortcuts persist across process death (managed by the system).

Core insight: **widgets run out-of-process via `RemoteViews` (Glance compiles to them), so render a glanceable snapshot of the SSOT (Ch 3) and push updates via WorkManager — periodic + on-change, respecting Doze/App Standby (Ch 47) and never over-refreshing; provide app shortcuts (static for fixed actions, dynamic for contextual/recent within the ~4–5 cap, pinned on user request) that, like widgets, deep-link (Ch 52) into the right destination with auth and account checks; and treat these as shared, always-visible surfaces — update them on account switch and keep sensitive data off them (Ch 42/28).**

55.4 High-Level Architecture



Components & ownership

- **SSOT** is the single data source; **WidgetUpdater** (WorkManager) pushes snapshots to Glance/`RemoteViews`.
- **Shortcuts** (static/dynamic/pinned) are launcher entry points.
- Both **deep-link** through the central router (Ch 52) with auth/account checks.
- **Account/privacy** logic keeps surfaces correct and safe.

Direction: SSOT + triggers → WorkManager update → widget snapshot; widgets/shortcuts → deep link → router → destination.

55.5 Detailed Component Design



Component	Responsibility
`OrderWidget` (Glance)	render SSOT snapshot as `RemoteViews`
`WidgetUpdateWorker`	read SSOT, push widget update (periodic + on-change, Ch 30/47)
`ShortcutPublisher`	manage dynamic/pinned shortcuts (cap-aware)
`WidgetDeepLinks`	build `FLAG_IMMUTABLE` deep-link PendingIntents (Ch 52)

55.6 End-to-End Data Flow

SSOT change → widget update



Widget/shortcut tap → deep link



55.7 Complete Kotlin Implementation

Package: `com.streamcart.widgets`.

55.7.1 Glance widget rendering an SSOT snapshot

```
class OrderWidget : GlanceAppWidget() {
    override suspend fun provideGlance(context: Context, id: GlanceId) {
        // Read a SNAPSHOT from the SSOT (Ch 3) for the ACTIVE account (Ch 42). No long-lived
logic.
        val entryPoint = EntryPointAccessors.fromApplication(context,
WidgetEntryPoint::class.java)
        val order = entryPoint.orderRepository().latestOrderSnapshot()   // suspend, from Room
        provideContent {
            Column(GlanceModifier.fillMaxSize().clickable(                // tap -> deep link
(Ch 52)

actionStartActivity(entryPoint.deepLinks().intent("streamcart://order/${order?.id}")))) {
                Text(order?.status ?: "No recent orders")                // glanceable, non-
sensitive
                Text(order?.etaText ?: "")
            }
        }
    }
}
class OrderWidgetReceiver : GlanceAppWidgetReceiver() { override val glanceAppWidget =
OrderWidget() }
```

55.7.2 WorkManager-driven updates (periodic + on-change, Ch 47)

```
@HiltWorker
class WidgetUpdateWorker @AssistedInject constructor(
    @Assisted ctx: Context, @Assisted p: WorkerParameters,
) : CoroutineWorker(ctx, p) {
    override suspend fun doWork(): Result = runCatching {
        OrderWidget().updateAll(applicationContext)    // re-reads SSOT snapshot, pushes
RemoteViews
    }.fold({ Result.success() }, { Result.retry() })
}

@Singleton
class WidgetUpdater @Inject constructor(private val wm: WorkManager) {
    fun schedulePeriodic() = wm.enqueueUniquePeriodicWork(          // respects Doze (Ch 47)
        "widget-refresh", ExistingPeriodicWorkPolicy.KEEP,
        PeriodicWorkRequestBuilder<WidgetUpdateWorker>(30, TimeUnit.MINUTES)

.setConstraints(Constraints.Builder().setRequiredNetworkType(NetworkType.CONNECTED).build()).bui
ld())

    fun updateNow() = wm.enqueue(OneTimeWorkRequestBuilder<WidgetUpdateWorker>()
        .setExpedited(OutOfQuotaPolicy.RUN_AS_NON_EXPEDITED_WORK_REQUEST).build())   // on data
change (Ch 47)

    fun cancel() = wm.cancelUniqueWork("widget-refresh")   // when last widget removed
(onDisabled)
}
```

55.7.3 App shortcuts: static (XML) + dynamic + pinned

```
<!-- Static shortcuts: res/xml/shortcuts.xml, referenced from the launcher activity meta-data --
>
<shortcuts xmlns:android="http://schemas.android.com/apk/res/android">
    <shortcut android:shortcutId="search" android:shortcutShortLabel="@string/sc_search"
              android:icon="@drawable/ic_search">
        <intent android:action="android.intent.action.VIEW"
                android:data="streamcart://search" android:targetPackage="com.streamcart"
                android:targetClass="com.streamcart.MainActivity" />
    </shortcut>
    <!-- "My Orders" similar -->
</shortcuts>
```

```
@Singleton
class ShortcutPublisher @Inject constructor(@ApplicationContext private val ctx: Context) {
    /** Dynamic: recent/contextual; capped (~4-5 incl. static). Update as context changes. */
    fun pushRecentSearches(terms: List<String>) {
        val max = ShortcutManagerCompat.getMaxShortcutCountPerActivity(ctx)
        terms.take((max - STATIC_COUNT).coerceAtLeast(0)).forEach { term ->
            val sc = ShortcutInfoCompat.Builder(ctx, "recent_$term")
                .setShortLabel(term)
                .setIntent(Intent(Intent.ACTION_VIEW, Uri.parse("streamcart://search?q=$term"),
ctx, MainActivity::class.java))
                .build()
            ShortcutManagerCompat.pushDynamicShortcut(ctx, sc)
        }
    }
    /** Pinned: only on explicit user action. */
    fun requestPin(id: String, label: String, deepLink: String) {
        if (!ShortcutManagerCompat.isRequestPinShortcutSupported(ctx)) return
        val sc = ShortcutInfoCompat.Builder(ctx, id).setShortLabel(label)
            .setIntent(Intent(Intent.ACTION_VIEW, Uri.parse(deepLink), ctx,
MainActivity::class.java)).build()
        ShortcutManagerCompat.requestPinShortcut(ctx, sc, null)
    }
    fun clearDynamic() = ShortcutManagerCompat.removeAllDynamicShortcuts(ctx)   // on logout (Ch
42)
}
```

55.7.4 Account switch + privacy

```
// On account switch (Ch 42): refresh widgets + dynamic shortcuts so they reflect the active
account.
suspend fun onAccountSwitched() {
    OrderWidget().updateAll(ctx)            // re-reads SSOT for the new active account
    shortcutPublisher.clearDynamic()        // clear previous account's contextual shortcuts
    // Privacy (Ch 28): widget shows non-sensitive summary; sensitive detail only inside the
app.
}
```

55.7.5 Module structure

```
:feature:widgets   OrderWidget (Glance), OrderWidgetReceiver, WidgetUpdateWorker, WidgetUpdater
:feature:shortcuts ShortcutPublisher (+ res/xml/shortcuts.xml)
:core:deeplink     WidgetDeepLinks / central router (Ch 52)
:core:data         SSOT repositories (Ch 3) via Hilt EntryPoint for the out-of-process widget
```

55.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Over-frequent updates	battery + rate limit	WorkManager cadence; update only on real change (Ch 47)
Update after widget removed	wasted work/crash	cancel work in `onDisabled` / `onDeleted`
SSOT read in launcher process	cross-process access	Hilt `EntryPoint` + Room (safe, read snapshot)
Account switch mid-update	stale/cross-account content	re-read active account (Ch 42); clear dynamic shortcuts
Dynamic shortcut cap exceeded	pushes dropped	respect `getMaxShortcutCountPerActivity`
Process death during update	partial	WorkManager retries; next update re-derives from SSOT

- **Update only when data changed:** enqueue a widget update on actual SSOT change (Ch 32) + a coarse periodic refresh — not on every event — to respect battery and update-rate limits (Ch 47).
- **Cancel update work when the last widget is removed** (`onDisabled`) so you don't keep refreshing a non-existent widget.
- **SSOT re-derivation makes widgets process-death-safe:** since the widget renders a snapshot read on each update, there's no widget state to restore — a benefit of SSOT-backing (Ch 3).

55.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Widget shows stale data	user report	update on SSOT change + periodic (Ch 30/47)	fresh
Battery drain from updates	vitals (Ch 47)	reduce cadence; update-on-change only	battery ok
Widget blank after reboot/death	logs	re-derive from SSOT on next update	recovers
Dynamic shortcuts empty/stale	user report	push/update as context changes; respect cap	current
Cross-account leak on widget	privacy review	update on switch; non-sensitive content	isolated
Tap opens wrong screen	QA	route via central deep-link router (Ch 52)	correct
Pinned shortcut deep link broken	QA	validate route; `FLAG_IMMUTABLE` PendingIntent	works



> **Widgets live or die by the SSOT + update cadence:** back them with the single source of truth (Ch 3) and update them on real changes plus a battery-respecting periodic job (Ch 30/47). Everything else — freshness, process-death resilience, account correctness — follows from rendering a snapshot of the SSOT rather than holding widget-local state.

55.10 Scaling the Implementation

- **Level 1.** A Glance widget rendering an SSOT snapshot; static shortcuts; deep-link taps.
- **Level 2.** WorkManager updates (periodic + on-change, Ch 47); dynamic shortcuts (contextual/recent); central deep-link routing (Ch 52).
- **Level 3 — production.** Multi-account updates (Ch 42) + privacy; cancel-on-remove lifecycle; pinned shortcuts on user action; battery-respecting cadence.
- **Level 4 — large.** Multiple widget types + sizes (responsive Glance); shortcut analytics (usage); per-account widget content; configurable widgets.
- **Level 5 — enterprise.** Widget/shortcut governance, engagement analytics (widget-driven opens), A/B of content, server-informed contextual shortcuts, consistent deep-link registry (Ch 52/53).

Avoid overengineering: most apps need one or two SSOT-backed widgets + a few static/ dynamic shortcuts + deep links. Don't build many rarely-used widgets or over-refresh them.

55.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Widget API	Glance (Compose-style)	classic `RemoteViews` / `AppWidgetProvider`	Glance for new (nicer API, same limits); classic for legacy
Widget data	SSOT-backed snapshot	widget-local state	SSOT always (fresh, process-death-safe)
Update cadence	WorkManager (on-change + periodic)	`updatePeriodMillis` only	WorkManager (reliable, finer, respects Doze); updatePeriodMillis is 30-min min + unreliable
Shortcut type	Static + dynamic (+ pinned on request)	all static	Mix: static for fixed, dynamic for contextual
Entry routing	Central deep-link router (Ch 52)	per-surface intent handling	Central (consistency, auth, account)
Sensitive content	Summary + tap into app	full detail on widget	Summary (privacy on shared surface)

Glance vs classic `RemoteViews`: Glance offers a Compose-style authoring experience and is the modern recommendation, but it **compiles to `RemoteViews`** and inherits all the out-of-process limits (limited components, pushed updates, no arbitrary logic). It doesn't remove the constraints — it just makes authoring pleasanter. Use Glance for new widgets; the execution model and SSOT/update requirements are identical to classic.

SSOT-backed vs widget-local data: holding data inside the widget leads to staleness, divergence from the app, and loss on process death. Rendering a **snapshot of the SSOT** (Ch 3) on each update keeps the widget consistent with the app, makes it process-death-safe (re-derived), and centralizes freshness in the sync layer (Ch 32). Always back widgets with the SSOT.

WorkManager updates vs `updatePeriodMillis`: the manifest `updatePeriodMillis` has a **30-minute minimum** and is deferred/unreliable under Doze — inadequate for timely updates. **WorkManager** (Ch 30) with constraints

(Ch 47) gives reliable, finer, battery- respecting updates and lets you push an **on-change** update promptly (expedited). Use WorkManager; treat `updatePeriodMillis` as a coarse backstop at most.

55.12 Security & Privacy Considerations

- **Widgets are an always-visible, potentially shared surface:** don't show **sensitive** content (balances, private messages, health) by default — show a non-sensitive summary and require opening the app (or an opt-in / tap-to-reveal) for detail (Ch 28).
- **Multi-account isolation (Ch 42):** update/clear widgets and dynamic shortcuts on account switch; never show account A's data under account B.
- **Deep-link safety (Ch 52):** widget/shortcut intents carry deep links — route through the central router that **validates params** and **authorizes server-side** (IDOR); build `PendingIntent`s with **`FLAG_IMMUTABLE`**; don't perform sensitive actions directly from a shortcut without in-app confirmation.
- **Exported receivers:** the `AppWidgetProvider` / `GlanceAppWidgetReceiver` is invoked by the system; validate inputs and keep logic minimal.
- **Shortcut data is visible:** dynamic shortcut labels (e.g., recent searches) appear on the launcher — don't put sensitive text there (Ch 28).
- Client vs backend: widgets/shortcuts render client-side SSOT data and deep-link in; the backend authorizes the destination like any entry point (Ch 52/23).

55.13 Performance and Resource Optimization

- **Update-on-change + coarse periodic** (Ch 47) is the main battery lever — don't poll or over-refresh.
- **Only update when data actually changed** (diff against last-pushed snapshot) to skip redundant `RemoteViews` pushes.
- **Keep widget content light:** small images (sized for the widget), minimal view tree — `RemoteViews` are serialized across processes.
- **Cancel work when no widgets remain** (`onDisabled`).
- **Respect the shortcut cap** (`getMaxShortcutCountPerActivity`) — extra pushes are dropped.
- **Profiling:** widget-update frequency + battery attribution (Battery Historian), widget/shortcut-driven app opens, update latency after SSOT change.

55.14 Testing Strategy

Glance has test utilities; shortcuts/WorkManager test with their APIs; deep links via Ch 52 patterns.

```
// Glance widget content (GlanceAppWidgetUnitTest / runGlanceAppWidgetUnitTest):
class OrderWidgetTest {
    @Test fun `renders latest order status from SSOT`() = runGlanceAppWidgetUnitTest {
        // provide a fake OrderRepository snapshot; assert the Text shows the status + deep-link
action
    }
}

class WidgetUpdateWorkerTest {
    @Test fun `updates widget from SSOT`() = runTest {
        val worker = TestListenableWorkerBuilder<WidgetUpdateWorker>(context).build()
        assertEquals(Result.success(), worker.doWork())   // (fake widget update)
    }
}

class ShortcutPublisherTest {
    @Test fun `respects max shortcut count`() { /* push N > cap -> only cap pushed */ }
    @Test fun `clearDynamic removes all on logout`() { /* Ch 42 */ }
}
```

What to test: widget renders correct SSOT snapshot + deep-link action; update worker pushes update; on-change + periodic scheduling (WM TestDriver); shortcut cap respected; dynamic shortcuts cleared on logout; account switch updates widget/shortcuts; deep-link tap routes correctly (Ch 52); no update after `onDisabled`. Instrumentation for real widget add/remove.

55.15 Production Observability

- **Metrics:** widget-update frequency + battery attribution (Ch 47); widget/shortcut- driven app-open rate (engagement); update latency after SSOT change; dynamic-shortcut usage; pinned-shortcut acceptance; stale-data reports; cross-account leak (0).
- **Logs:** `widget{type, trigger, updated, account}` / `shortcut{type, id, launched}` — no sensitive content/PII (Ch 28).
- **Debugging:** stale widget → update not triggered on SSOT change / Doze deferral; battery → over-refresh; blank → process death + no re-derive; wrong screen → deep-link routing (Ch 52); empty shortcuts → not pushed / cap exceeded.
- **Alerting:** widget-update battery spike; cross-account leak; deep-link failure from these surfaces.
- **Dashboards:** widget/shortcut engagement (opens driven); update cost vs freshness.

55.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *How does a widget run?* Out-of-process in the launcher via `RemoteViews`; updates pushed from your process.
2. *Glance?* Compose-style widget API that compiles to `RemoteViews` — same limits.
3. *Can a widget run arbitrary code?* No — limited view tree, pushed updates.
4. *Where does widget data come from?* The SSOT (Ch 3) — render a snapshot.
5. *Process death?* Widget re-derives from SSOT on next update (no state to restore).

Updates

6. *How to update a widget?* WorkManager (periodic + on-change) pushing new `RemoteViews`.
7. `updatePeriodMillis` *limits?* 30-min minimum, unreliable under Doze.
8. *Respect background limits?* Yes — cadence per Doze/App Standby (Ch 47); update-on- change.
9. *Over-update cost?* Battery + rate limits.

Shortcuts

10. *Three types?* Static (manifest, fixed), dynamic (runtime, contextual), pinned (user- placed).
11. *Cap?* ~4–5 visible (`getMaxShortcutCountPerActivity`).
12. *Dynamic use?* Recent/contextual; update as context changes.
13. *Pinned?* Request only on user action.
14. *What do they fire?* An Intent / deep link (Ch 52).

Deep links

15. *Routing?* Central router (Ch 52): destination + back stack + auth + account.
16. *PendingIntent?* `FLAG_IMMUTABLE`.
17. *Untrusted?* Yes — validate + server authz (Ch 52/23).

Multi-account/privacy

18. *Account switch?* Update/clear widgets + dynamic shortcuts (Ch 42).
19. *Sensitive content on widget?* Avoid — summary only; detail in-app (Ch 28).
20. *Shortcut labels?* No sensitive text (visible on launcher).

Lifecycle

21. *Widget removed?* `onDisabled` / `onDeleted` → cancel update work.
22. *SSOT access from widget?* Hilt `EntryPoint` + Room snapshot.

Testing/Production

23. *Test widget content?* Glance unit test utilities.
24. *Test shortcuts?* Cap + clear-on-logout.
25. *Stale widget cause?* Update not triggered / Doze deferral.
26. *Battery?* Over-refresh — reduce cadence.

Design-change challenges

1. **Order-status widget that stays fresh.** SSOT-backed + WorkManager on-change + periodic (Ch 30/47).
2. **Long-press quick actions (Search, My Orders, recent).** Static + dynamic shortcuts → deep links.
3. **Widget drains battery.** Update-on-change only + coarse periodic; respect Doze.
4. **Widget shows the wrong account after switching.** Update on switch; non-sensitive content (Ch 42/28).
5. **Resume last show from the launcher.** Dynamic shortcut + widget tile deep-linking to the player (Ch 58).

Mock walkthrough (abridged)

Clarify (what content, how fresh, which actions, shared surface/sensitive?) → widget = `RemoteViews`/Glance snapshot of SSOT (Ch 3) → updates via WorkManager (on-change + periodic, Ch 47) → shortcuts (static/dynamic/pinned, cap) → deep-link both via central router (Ch 52, auth/account) → multi-account + privacy (Ch 42/28) → lifecycle (cancel on remove) → testing/observability (engagement, battery).

55.17 Practical Exercises

1. **Beginner.** Build a Glance widget showing the latest order status from the SSOT, with a deep-link tap. *Eval:* renders snapshot; tap opens the order.
2. **Intermediate.** Update the widget on SSOT change and on a periodic WorkManager job respecting Doze; cancel work when the last widget is removed. *Eval:* fresh; battery- respecting; cancels.
3. **Advanced.** Add static shortcuts (Search, My Orders) and dynamic recent-search shortcuts (cap-aware) that deep-link; clear dynamic on logout. *Eval:* correct types; cap respected; cleared.
4. **Production debugging.** Fix a stale widget (update not triggered on sync) and battery drain (over-refresh). *Eval:* updates on change; cadence reduced.
5. **System design challenge.** Make widgets + dynamic shortcuts account-aware (update on switch, no leak) and add engagement analytics. *Eval:* account-isolated; opens tracked.

(Solutions withheld unless requested.)

55.18 Chapter Summary & Reference Sheet

Key decisions: build widgets with **Glance** (which compiles to `RemoteViews` and runs **out-of-process** in the launcher, so no arbitrary logic — updates are **pushed**), rendering a **snapshot of the SSOT** (Ch 3) so they're fresh and process-death-safe; **update via WorkManager** (**on-change + a coarse periodic job**, respecting Doze/App Standby, Ch 30/47) rather than the unreliable 30-minute `updatePeriodMillis`, and only when data actually changed; provide **app shortcuts** — **static** (fixed, manifest), **dynamic** (contextual/recent, within the ~4–5 cap), **pinned** (on user request) — that **deep-link** (Ch 52) into the app; route **all** widget/shortcut taps through the **central deep-link router** (destination + back stack + **auth** + **account**, with `FLAG_IMMUTABLE` PendingIntents); and treat these as **shared, always-visible surfaces** — **update/clear on account switch** (Ch 42) and keep **sensitive content off** them (summary in the widget, detail in the app, Ch 28).

Concepts: `RemoteViews`/out-of-process widgets, Glance, SSOT-backed snapshots, WorkManager updates (on-change + periodic) under Ch 47, app shortcuts (static/dynamic/ pinned + cap), deep-link integration, multi-

account/privacy, widget lifecycle.

Common mistakes: treating a widget like an Activity (arbitrary logic); widget-local data (stale/lost); `updatePeriodMillis`-only updates (unreliable/coarse); over-refreshing (battery); wrong shortcut type / ignoring the cap; not updating on account switch (leak/ wrong data); sensitive content on the widget; not routing taps through the central deep- link router; not cancelling work when widgets are removed.

Checklist:

- [ ] Glance/ `RemoteViews` snapshot of SSOT (Ch 3)
- [ ] WorkManager updates (on-change + periodic, Ch 47)
- [ ] update only on real change
- [ ] static/dynamic/pinned shortcuts (respect cap)
- [ ] deep-link via central router (Ch 52) + `FLAG_IMMUTABLE`
- [ ] auth/account checks
- [ ] update/clear on account switch (Ch 42)
- [ ] no sensitive content on widget (Ch 28)
- [ ] cancel work on `onDisabled`
- [ ] engagement/battery telemetry.

Key APIs: `GlanceAppWidget` / `GlanceAppWidgetReceiver` / `provideContent` / `updateAll` / `updateAppWidgetState`, `AppWidgetProvider` / `AppWidgetManager` / `RemoteViews`, `ShortcutManagerCompat` (`pushDynamicShortcut` / `requestPinShortcut` / `getMaxShortcutCountPerActivity` / `removeAllDynamicShortcuts`), `ShortcutInfoCompat`, static shortcuts XML, `CoroutineWorker` /WorkManager (Ch 30), Hilt `EntryPoint`, `PendingIntent` / `actionStartActivity` (`FLAG_IMMUTABLE`, Ch 52).

References:

- App widgets / Glance: https://developer.android.com/develop/ui/compose/glance
- Create a widget (RemoteViews): https://developer.android.com/develop/ui/views/appwidgets/overview
- App shortcuts: https://developer.android.com/develop/ui/views/launch/shortcuts
- Update widgets with WorkManager: https://developer.android.com/develop/ui/compose/glance/glance-app-widget
- (composes) Ch 3 SSOT, Ch 30 WorkManager, Ch 47 Doze/Limits, Ch 52 Deep Links, Ch 42 Multi-Account, Ch 28 Privacy

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; Glance widget + update worker/scheduler + shortcuts (static/dynamic/pinned) + account/ privacy code; concurrency (update-on-change, cancel-on-remove, SSOT re-derive); failure (stale/battery/leak triage); 5 levels; Glance-vs-RemoteViews, SSOT-backing, WorkManager- vs-updatePeriodMillis trade-offs; security (shared surface, account, deep-link safety); perf (update-on-change); Glance/WM/shortcut tests; observability (engagement/battery); 26 Q&A + 5 challenges. ✅ Completes Part VI.
- **Expand later:** deep links (Ch 52), WorkManager (Ch 30), Doze (Ch 47), media (Ch 58).
- **Interview coverage:** exceeds ≥25 bar.